

Migration Copilot

AI-assisted migration of legacy ETL (Informatica, SSIS, Talend, DataStage) into native Pentaho Data Integration pipelines.

The opportunity

Every legacy ETL platform locks customers in the same way — not with superior technology, but with the sunk cost of thousands of hand-built pipelines. A large enterprise commonly runs 3,000–10,000 mappings, and rebuilding them by hand to change vendors is a multi-year, seven-figure engagement. That switching cost, not product capability, is the main reason organizations stay on tools they have outgrown.

The Migration Copilot turns that migration from an 18-month services project into an assisted effort measured in weeks. In doing so it removes the single largest barrier to adopting Pentaho — converting migration from a cost the sales team apologizes for into the reason a prospect switches.

How it works

The system is a four-stage pipeline. The central design principle is that not every stage should be AI: the reliable product is a hybrid that uses deterministic parsing where accuracy is non-negotiable and applies AI only where semantic judgment is genuinely required.

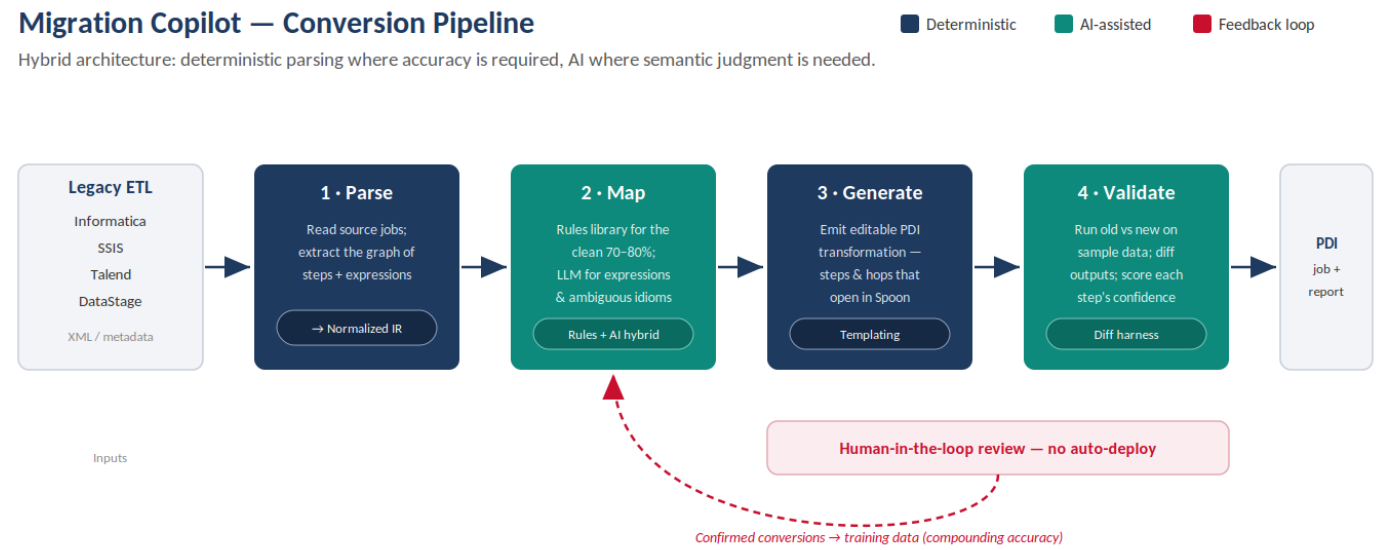


Figure 1 — Conversion pipeline. Navy = deterministic, teal = AI-assisted.

- PARSE** Deterministic. Source jobs are structured XML or metadata, so real parsers extract the pipeline graph — sources, targets, transformations, expressions — into a normalized intermediate representation. No hallucination risk.
- MAP** Hybrid. A rules library handles the clean 70–80% (an Informatica Aggregator → PDI Group By, a Lookup → Stream Lookup). The LLM handles the messy remainder: translating proprietary expression languages and resolving idioms with no 1:1 equivalent.
- GENERATE** Emit a valid, editable PDI transformation — steps and hops that open cleanly in Spoon so an engineer can refine, never a black box.
- VALIDATE** Run the original and converted pipeline against sample data, diff the outputs, and score each step's confidence. The migration report flags what was auto-converted, what is a best-guess needing review, and what a human must handle.

Where AI adds value — and where it must not

Stage	Approach	Rationale
Parse	Deterministic parsers	Source formats are structured; accuracy is required and achievable without AI.
Map (clean 70–80%)	Rules library	Known 1:1 step equivalents; faster and safer as fixed mappings.
Map (messy 20–30%)	LLM translation	Expression-language and idiom translation is exactly what LLMs do well.
Generate	Templating from IR	Deterministic emission of valid, editable PDI artifacts.

Stage	Approach	Rationale
Validate	Diff harness + scoring	Output-level safety net; catches subtly wrong conversions before humans do.

Phased roadmap

Phase	Scope	Why this order
0 — Internal tool	Informatica PowerCenter only; top ~20 transformation types; used by Pentaho's own services team.	De-risks accuracy on real jobs and generates training data before any customer sees it.
1 — Assisted product	Same source, exposed to customers with confidence scoring and mandatory human review.	Sells the 80% time saving with an honest map of the remaining 20%.
2 — Multi-source	Add SSIS, then Talend / DataStage; broaden transformation coverage.	Expands the addressable displacement market once the core engine is proven.

Discipline matters more than coverage: the tool should handle the common 80% flawlessly and cleanly hand off the rest, rather than chase every customer-specific one-off.

Key risks & mitigations

Risk	Mitigation
Subtly wrong conversions (worse than obvious failures in ETL)	Output-diffing validation harness + per-step confidence scoring; no auto-deploy.
Expression-language translation errors	Constrain LLM with the rules library and validated examples; flag low-confidence translations for review.
Proprietary / heavily customized source components	Detect and route to human handoff rather than guess; report them explicitly.
Scope creep across environments	Fixed supported-feature list per phase; explicit unsupported-feature reporting.

Business case

Three returns stack. As a sales weapon it neutralizes "switching is too expensive" and shortens the deal cycle. As a services accelerator it makes Pentaho's own migration engagements far more profitable — or wins fixed-price bids competitors cannot. And it builds a data moat: every migration teaches the mapping models, so accuracy compounds with adoption in a way competitors cannot easily replicate.

Success metrics

Hours-to-migrate per job versus the manual baseline; percentage of steps auto-converted without human edits; validation pass rate (output parity old vs. new); and, commercially, reduction in migration-driven deal-cycle length.